

Dual-Channel Adaptive Scale Hypergraph Encoders With Cross-View Contrastive Learning for Knowledge Tracing

Jiawei Li¹, Yuanfei Deng¹, Yixiu Qin¹, Shun Mao¹, and Yuncheng Jiang¹, *Member, IEEE*

Abstract—Knowledge tracing (KT) refers to predicting learners’ performance in the future according to their historical responses, which has become an essential task in intelligent tutoring systems. Most deep learning-based methods usually model the learners’ knowledge states via recurrent neural networks (RNNs) or attention mechanisms. Recently emerging graph neural networks (GNNs) assist the KT model to capture the relationships such as question–skill and question–learner. However, non-pairwise and complex higher-order information among responses is ignored. In addition, a single-channel encoded hidden vector struggles to represent multigranularity knowledge states. To tackle the above problems, we propose a novel KT model named dual-channel adaptive scale hypergraph encoders with cross-view contrastive learning (HyperKT). Specifically, we design an adaptive scale hyperedge distillation component for generating knowledge-aware hyperedges and pattern-aware hyperedges that reflect non-pairwise higher-order features among responses. Then, we propose dual-channel hypergraph encoders to capture multigranularity knowledge states from global and local state hypergraphs. The encoders consist of a simplified hypergraph convolution network and a collaborative hypergraph convolution network. To enhance the supervisory signal in the state hypergraphs, we introduce the cross-view contrastive learning mechanism, which performs among state hypergraph views and their transformed line graph views. Extensive experiments on three real-world datasets demonstrate the superior performance of our HyperKT over the state-of-the-art (SOTA).

Index Terms—Contrastive learning, hyperedge distillation, hypergraph learning, knowledge tracing (KT).

NOMENCLATURE

$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Hypergraph $\mathcal{G}$ with the node set $\mathcal{V}$ and edge set $\mathcal{E}$.
$\mathbf{H} \in \mathbb{R}^{ \mathcal{V} \times \mathcal{E} }$	Incidence matrix.
$L(\mathcal{G})$	Line graph of hypergraph $\mathcal{G}$.
$\mathcal{E}^k$	Set of knowledge-aware hyperedges.

Manuscript received 27 April 2023; revised 27 December 2023; accepted 2 April 2024. Date of publication 23 April 2024; date of current version 7 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 61772210 and Grant U1911201, in part by the Key Research and Development Program of Guangdong of China under Grant 2023B0303010004, and in part by the Innovation Team Project for Universities in Guangdong Province in China under Grant 2023KCXTD011. (Corresponding author: Yuncheng Jiang.)

Jiawei Li, Yuanfei Deng, Yixiu Qin, and Shun Mao are with the School of Computer Science, South China Normal University, Guangzhou 510631, China (e-mail: lijiawei@m.scnu.edu.cn; dengyf@m.scnu.edu.cn; yxqin@m.scnu.edu.cn; shunm@m.scnu.edu.cn).

Yuncheng Jiang is with the School of Computer Science, South China Normal University, Guangzhou 510631, China, and also with the School of Artificial Intelligence and the School of Software, South China Normal University, Foshan 528225, China (e-mail: ycjiang@scnu.edu.cn).

Digital Object Identifier 10.1109/TNNLS.2024.3386810

$\mathcal{E}^p$	Set of pattern-aware hyperedges.
$\mathbf{H}^k \in \mathbb{R}^{ \mathcal{V} \times \mathcal{E}^k }$	Knowledge-aware hyperedges incidence matrix.
$\mathbf{H}^p \in \mathbb{R}^{ \mathcal{V} \times \mathcal{E}^p }$	Pattern-aware hyperedges incidence matrix.
U	Set of learners.
Q	Set of questions.
X	Set of learner’s historical responses.
$s(q_a, q_b)$	Similarity of question q_a and question q_b .
f	Function of knowledge-aware hyperedge generation.
g	Function of pattern-aware hyperedge generation.
$\mathcal{P}_c(X)$	Set of all consecutive subsequences of X .
γ	Intervention parameter of question similarity.
ρ	Learning pattern-aware control function.
$\mathbf{X}^{(l)}$	Embeddings of nodes at layer l in hypergraph $\mathcal{G}$.
$\Psi^{(l)}$	Embeddings of nodes at layer l in line graph $L(\mathcal{G})$.
θ	Embeddings of knowledge state.

I. INTRODUCTION

WITH the rapid expansion of online intelligent tutoring systems, there has been an exponential increase in the abundance of question resources and learner behavior data [1]. Assessing these data proves instrumental in enabling learners to accurately understand their strengths and weaknesses, consequently facilitating the development of personalized learning paths [2]. As an effective technology for educational data mining, knowledge tracing (KT) lays the foundation for human cognitive states tracing, adaptive learning systems, and exercise recommendations [3], [4]. The KT model aims to accurately predict learners’ future performance by tracing the knowledge mastery states [5], [6]. Specifically, the KT model predicts the probability of learners correctly answering the next question based on their historical response records.

The existing KT models based on Bayesian [7] and factor analysis [8] represent typical methods that have consistently demonstrated strong performance. Leveraging the powerful representation capabilities of deep learning, there is a discernible trend toward KT research based on deep learning methods [9], [10]. To acquire the hidden knowledge state, early

deep learning-based KT approaches encode learners' response sequences via recurrent neural networks (RNNs) [11], memory networks [12], and attention mechanism [9]. Recent studies have focused on modeling diverse learning behaviors from response sequences, which are beneficial to improving the performance of the KT model, such as the forgetting mechanism [13], [14], learner individualization [15], temporal cross-effects [16], and learning gains [14].

While optimizing the modeling of the learning process and learning behavior can enhance prediction performance, a significant challenge arises from the sparse interaction between learners and a large number of questions [17]. To address the data sparsity issue, some approaches aim to incorporate textual information of questions into the KT model [6]. However, the textual content of questions is often challenging to obtain [10]. Recently, graph neural networks (GNNs) have demonstrated notable success in modeling graph structural data [18]. Graph-based KT models have exhibited remarkable performance, utilizing message-passing on question-skill graphs to refine question embeddings [19], [20]. Other graph-based KT methods focus on modeling latent knowledge states from learner-question interaction graphs [21].

However, a significant limitation of existing KT models is that they only utilize pairwise relationships to model knowledge states, overlooking higher-order relationship patterns prevalent among responses [22]. For instance, the non-pairwise high-order relationship among responses of interacting subsequence reflects the learner's knowledge state within that time span. Unlike graph-based models that emphasize pairwise relationships, hypergraph generalizes edges to connect more than two nodes [23], [24], offering a natural manner to capture complex higher-order relationships among responses. While the incorporation of hypergraphs to model higher-order relationships among responses is advantageous for inferring learners' knowledge states, generating effective hyperedges poses a challenge [25]. In addition, existing KT methods typically encode the knowledge state in a single channel [26], resulting in the inability to model the multigranular knowledge states during the learning process. In fact, response sequences under different views represent diverse knowledge states of learners.

To address the above issues, this article proposes dual-channel adaptive scale hypergraph encoders with cross-view contrastive learning for KT (HyperKT), which consists of three key learning components. First, to capture the non-pairwise and complex high-order relationships among responses, we propose a hyperedge distillation component for constructing state hypergraphs. Leveraging prior knowledge of problem skills and learning patterns during the learning process, hyperedge distillation adaptively extracts knowledge-aware hyperedges and pattern-aware hyperedges from response sequences. A distillation matrix is obtained through hyperedge distillation, facilitating the capture of essential information transmission between nodes and hyperedges in the hypergraph. Then, we construct global and local state hypergraphs, which are encoded via introduced dual-channel hypergraph encoders to capture multigranularity knowledge states. One channel of the hypergraph

encoder employs a simplified hypergraph convolution network, efficiently modeling learners' global knowledge states. In particular, for the local state hypergraph encoder channel, we propose a collaborative hypergraph convolution network to model fine-grained local knowledge states, with collaborative information comprising hyperedge distillation matrices and attention weight matrices. Finally, to address the issue of insufficient supervision signals in the state hypergraph caused by the sparsity of response sequences, we design a cross-view hypergraph contrastive learning mechanism, performing local and global contrastive learning between two hypergraph views and their transformed line graph views. The cross-view contrastive learning on the state hypergraphs maximizes the mutual information between the learned local knowledge state and the global knowledge state.

The main contributions of the proposed HyperKT are summarized as follows.

- 1) To capture non-pairwise high-order relationships among responses, we propose a hyperedge distillation component to guide the generation of knowledge-aware and pattern-aware hyperedges from response sequences.
- 2) We design dual-channel hypergraph encoders to model multigranular knowledge states from global and local state hypergraphs, including simplified and collaborative hypergraph convolution channels.
- 3) To enhance self-supervised signals in hypergraphs, we introduce a cross-view contrastive learning mechanism, maximizing the mutual information of the global and local knowledge states learned from multiviews.
- 4) Extensive experiments on three public datasets demonstrate that the proposed HyperKT outperforms various state-of-the-art (SOTA) models.

The remainder of this article is organized as follows. Section II reviews the relevant works of this article. Sections III and IV describe the problem definition and the details of the proposed HyperKT. The experimental results and analyses are presented in Section V. Section VI concludes this article.

II. RELATED WORKS

A. Knowledge Tracing

The goal of KT is to trace learners' changing knowledge states during exercise and correctly predict their performance on the subsequent questions. Existing KT methods can be categorized into traditional machine learning-based methods and deep learning-based methods.

Traditional machine learning-based Bayesian KT (BKT) [27] models the learner's knowledge state as a binary variable and updates it using Bayesian principles. Dynamic BKT (DBKT) [28] models the complicated relationship between skills using dynamic Bayesian networks. Factorization machines are also a tool for traditional machine learning-based methods [29]. KT machines (KTM) [8] use factoring machines to provide supplementary information to the KT model to improve performance.

Deep KT (DKT) [11], the first study to introduce deep learning in KT, models learners' knowledge states via RNNs. The

introduction of the attention mechanism further improves the performance of the KT model [9], [30]. Learners' performance on the future question is also affected by other elements, such as complicated learning behaviors. DKT-forgetting [13] uses three types of forgotten information as input characteristics. Hawkes Knowledge Tracing (HawkesKT) [16] models temporal cross-effects adaptively in the learning process. To raise learners' performance prediction, cognitive diagnostics [31] and internal mining relationships of learners [15] are also effective solutions.

However, data sparsity is a major issue for KT tasks. Thus, exercise-aware KT (EKT) [6] learns the semantic representation of each exercise from the text content to enhance KT model performance. Unfortunately, the content of the question is hard to obtain [21]. The graph-based KT models utilize the question-skill bipartite graph to help solve the sparse interactive data issue [32]. The graph-based interaction model for KT (GIKT) [19] learns the representation of the questions via the higher-order link between questions and skills. Another graph-based model combines gating-controlled forgetting and learning mechanisms (GFLDKT) [20] to model learners' knowledge states. Even though the above KT method shows excellent performance, it is still unable to model non-pairwise and complex higher-order information among responses. We leverage the hypergraph's ability to model high-order features and propose dual-channel hypergraph encoders to capture multigranular knowledge states from constructed state hypergraphs.

B. Hypergraph Learning

In recent years, there has been a rise in interest in hypergraph learning as a result of its adaptability and aptitude in modeling complicated data correlation [24]. Hypergraph learning has many real-world applications, such as recommendation system [23], link prediction [25], and computer vision [18]. Hypergraph convolutional networks (HyperGCN) [33] is the first method to apply graph convolution to hypergraphs, which is a generalization of GNNs. Hypergraph attention network (HGAT) [34] assists hypergraph general message passing in a more flexible way. Gatta et al. [23] proposed a hypergraph embedding method for music recommendation (HEMR), which generates embeddings from hypergraph data structures more efficiently. CentSmoothie [35] formulates drug-drug interaction prediction as a hypergraph, employing a novel "central-smoothing" approach to jointly learn representations of drugs and labels.

The effectiveness of hypergraph learning is influenced directly by the quality of the generated hyperedge [25]. The existing hyperedge generation is divided into implicit and explicit categories [24]. Implicit methods generate per-topic representations and metrics to characterize sample connections [36]. The explicit methods generate hyperedges via inherent structural information of data [37]. Designing an appropriate method to generate efficient hyperedges from complex learner response sequences is a challenge.

Inspired by the work of Xia et al. [38] and Li et al. [39], we construct state hypergraphs to model learner response sequences on the KT task, filling this research gap. Among

the aforementioned methods, HyperGCN, HGAT, HEMR, and CentSmoothie may encounter difficulties in capturing the rich information embedded within complex hypergraph structures, primarily due to the limitations of single-channel feature encoding. Therefore, we introduce dual-channel hypergraph encoders for modeling multigranularity knowledge states and design an adaptive scale hyperedge distillation component for hyperedge generation.

C. Contrastive Learning

Contrastive learning has achieved great success in self-supervised representation, which applies to computer vision [40] and natural language processing [41]. The latest advances in this area have extended contrastive learning to graph representation learning [42], [43]. The contrastive learning method learns node representations by contrasting positive pairs against negative pairs [44]. In other words, contrastive learning brings together similar representations and separates dissimilar ones. Finally, utilizing a contrastive learning objective maximizes agreement between two augmented views of the same interaction sequence.

Contrastive learning-based research in KT is currently in its early stages [17], [45]. Bigraph contrastive learning-based KT (Bi-CLKT) [46] augments the input graph by randomly removing edges and masking node characteristics in the "exercise-to-exercise" and "concept-to-concept" graphs. A contrastive learning framework for KT (CL4KT) [45] generates the augmentation sequence from the original response with various data augmentation methods as negative samples in contrastive loss. Despite the successful application of contrastive learning to KT tasks, it predominantly involves random data augmentation, indicating that the full potential of contrastive learning in this context has not been thoroughly explored. To improve the effectiveness of graph contrastive learning, an adaptive sampling strategy (AdaS) [47] enhances model performance by dynamically encoding the importance of negative nodes and mitigating the impact of false negatives through an auxiliary polarization regularizer. Zhao et al. [48] introduced a structure-adaptive graph contrastive learning framework, incorporating a structure learning layer and denoising supervision mechanism to promote structure learning of enhanced graphs. To address the diminishing utility of augmented views, Zhuo and Tan [49] presented a view update scheme that dynamically adjusts the augmented view, ensuring a continuous supply of novel information crucial for effective contrastive learning process.

While these methods have improved the generation of augmented views, there is a potential risk of losing critical information. We design a cross-view hypergraph contrastive learning module in HyperKT that employs line graphs to transform local and global state hypergraphs, ensuring consistency and generalization across different views.

III. PRELIMINARY AND PROBLEM FORMULATION

In this section, we first formally define the problem of learner performance prediction in the KT task. For clarification, we summarize necessary notations and their descriptions throughout this article in Nomenclature.

The KT task is to monitor learners' changing knowledge state during the learning process and predict learners' performances on future questions.

Definition 1 (Knowledge Tracing): We formalize the learner's learning process: Given a set of learners $U = \{u_1, u_2, \dots, u_n\}$, a set of questions $Q = \{q_1, q_2, \dots, q_m\}$, and a record of learners' historical question responses $X = \{(q_1, a_1), \dots, (q_t, a_t)\}$, where a_t is the correctness of the learner's response to the question q_t at timestep t

$$a_t = \begin{cases} 1, & \text{if the learner's response is right} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The objective of KT is to predict the probability of the learner answering the exercise correctly at the next learning step $t + 1$, i.e., y_{t+1} , which can be further applied to individualize learners' learning schemes and increase their learning effectiveness.

Then, we briefly explain the necessary definitions of hypergraphs, which are the base component of our model.

Definition 2 (Hypergraph): Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a hypergraph, where $\mathcal{V}$ is a set of vertex, and $\mathcal{E}$ is a set of hyperedge. Each hyperedge $e \in \mathcal{E}$ contains two or more vertices. The hypergraph can be represented by an incidence matrix $\mathbf{H} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{E}|}$.

Definition 3 (Line Graph of Hypergraph): Let $L(\mathcal{G})$ denote the line graph of the hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each hyperedge of $\mathcal{G}$ is a node in $L(\mathcal{G})$. If two hyperedges share at least one common vertex in $\mathcal{G}$, there is an edge among their corresponding two nodes in $L(\mathcal{G})$.

To transform the responses into hypergraphs, we need to define the hyperedges to capture the multigranularity knowledge state. At the same time, we define hyperedge distillation to control the generation of hyperedges, which can reduce the noise interference in hyperedges.

Definition 4 (Knowledge-Aware Hyperedge): Establishing connections among questions through prior question-skill relationships is beneficial for improving the performance of the KT model [32]. For example, for the interaction fragment $x_i = \{q_{t_1}, q_{t_2}, q_{t_3}, q_{t_4}\}$, assuming that the similarity between q_{t_1} and q_{t_4} (calculated from prior question-skill bipartite graph via graph embedding methods) is greater than the threshold γ , a hyperedge connecting all question nodes in x_i is defined as knowledge-aware hyperedge e^k . This kind of hyperedge reveals the flow of knowledge in response intervals.

Definition 5 (Pattern-Aware Hyperedge): The short-term responses in the interaction sequence reflect learners' immediate learning patterns [14], such as impatience and lack of attention during exercises, leading to inconsistencies between learners' performance and their actual knowledge states. For instance, for an interaction segment $x_j = \{(q_{t_1}, a_{t_1}), (q_{t_2}, a_{t_2}), \dots, (q_{t_w}, a_{t_w})\}$, if the correctness rate of this segment is too low (inconsistent with the learner's overall learning process), it indicates that the learner may not have invested enough effort or attempted questions beyond their grasp of knowledge. Therefore, the interaction segment of size w reflects the learner's learning pattern, and the questions in x_j are connected to form hyperedge, defined as pattern-aware hyperedge e^p .

Definition 6 (Hyperedge Distillation): Following [25], we define hyperedge distillation as the process of automatically generating adaptive scale hyperedges from responses. This process includes the standardization of knowledge-aware hyperedges and pattern-aware hyperedges, avoiding interference from noise hyperedges in an adaptive scale manner.

IV. METHODS

A. Overview

The overall framework of HyperKT is shown in Fig. 1, which comprises three main modules.

- 1) *Adaptive scale hyperedge distillation:* It generates hyperedges from learners' historical responses in an adaptive scale manner, including knowledge-aware hyperedges and pattern-aware hyperedges. These hyperedges are driven by the knowledge in the question and learning instant patterns. Hyperedge distillation assists HyperKT to capture fine-grained knowledge states.
- 2) *Dual-channel hypergraph encoders:* It constructs two state hypergraph views for learners, global state hypergraph and local state hypergraph. They are encoded by the proposed dual-channel hypergraph convolution networks to learn multigranularity knowledge states.
- 3) *Cross-view hypergraph contrastive learning:* It regards the hyperedges in the state hypergraphs as nodes to construct the corresponding line graphs to enhance the self-supervised signal. Mutual information between the learned multigranularity knowledge states is maximized by performing cross-view contrastive learning between state hypergraphs and line graphs. We next present the three modules in detail.

B. Adaptive Scale Hyperedge Distillation

As mentioned in Section I, the performance of learners is determined by the combined effect of non-paired responses. The modeling of higher-order information among these responses can be achieved through constructed hypergraphs. Hence, the proficient formation of hyperedges constitutes a fundamental component of KT based on hypergraphs. However, it remains challenging to extract the hyperedges that implicate the latent knowledge state from the response sequences. The straightforward approach is to intercept the response sequence with a fixed-size window to construct hyperedges, which may introduce interference or overlook critical information. Fortunately, the questions in the response sequences can be linked through the prior question-skill bipartite graph [32]. In addition, different fragments in response sequences reflect different learning patterns of learners [14]. Therefore, we propose a hyperedge distillation module to adaptively extract knowledge-aware hyperedges and pattern-aware hyperedges from response sequences by leveraging prior knowledge of question-skill relationships and learning patterns during the learning process.

Specifically, this module enables joint modeling of knowledge flow intervals and learning pattern intervals across adaptive scales. Responses within the interval that starts and ends with similar questions reflect the change in learners'

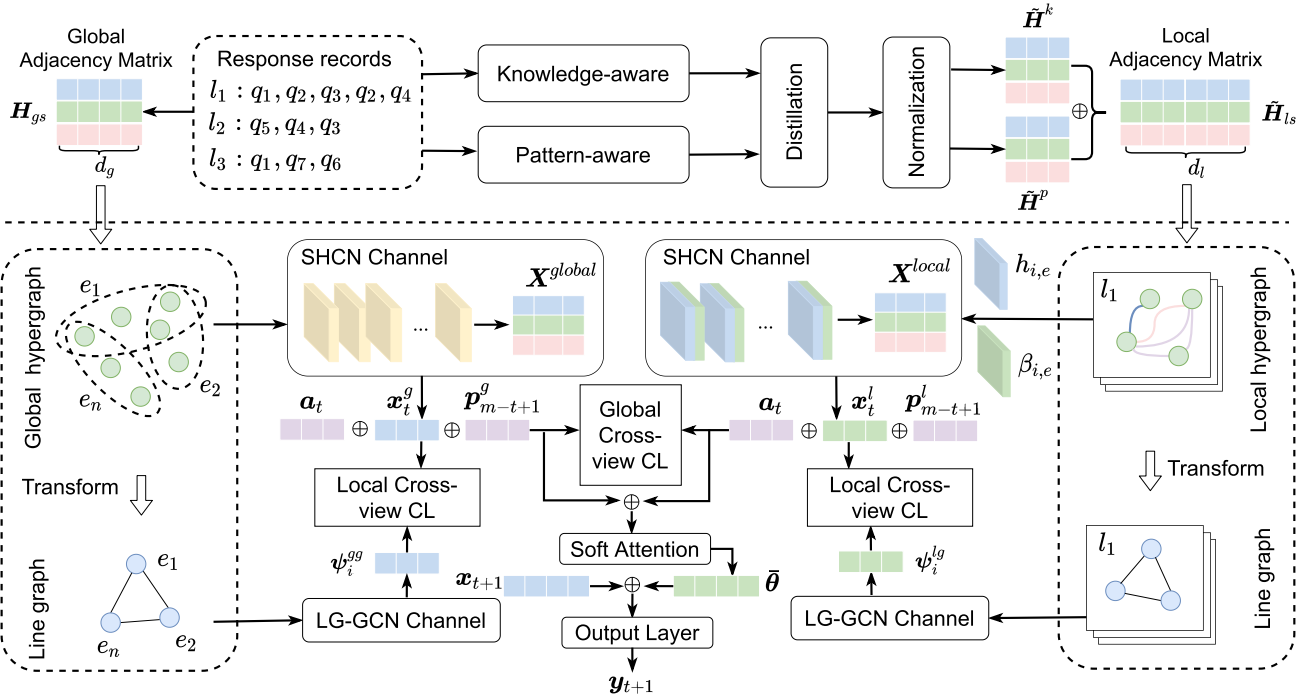


Fig. 1. Overall architecture of HyperKT. The top is the adaptive scale hyperedge distillation process, and the details of the distillation are given in the subsequent figures. The middle SHCN channel and CHCN channel are simplified hypergraph convolution networks and collaborative hypergraph convolution networks, respectively. Note that $h_{i,e}$ and $\beta_{i,e}$ are collaborative factors, representing hyperedge distillation weights and hyperedge attention weights.

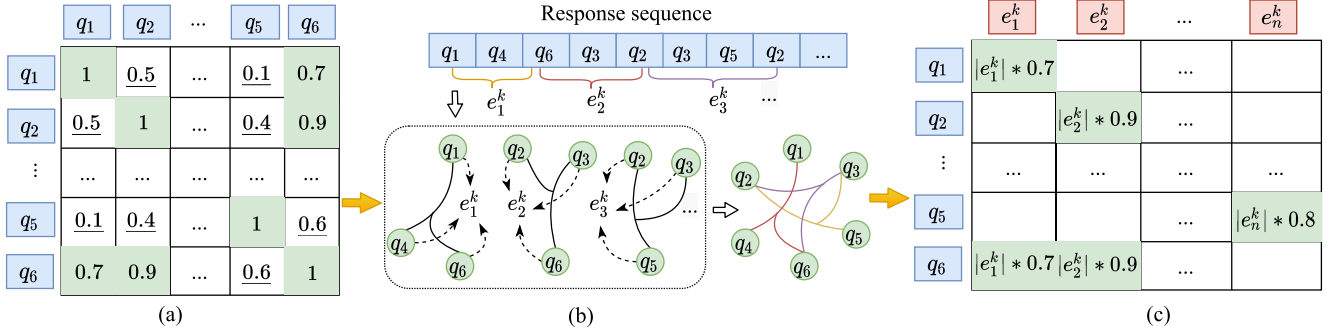


Fig. 2. Hyperedge distillation with knowledge-aware. We first calculate (a) similarity matrix of questions; then (b) generate knowledge-aware hyperedges in the response sequence based on question pairs with higher correlation (green marked in similarity matrix of questions); and finally, (c) incidence matrix is generated based on the length of the hyperedges and the similarity value of the question pairs.

question-solving capabilities. For example, a too-long interval may lead to the forgetting of pertinent knowledge, while a growing trend in the correct rate over the interval indicates the positive state of learners. Therefore, as shown in Fig. 2, we generate adaptive scale hyperedges with knowledge-aware. First, we construct a bipartite graph based on the prior knowledge that questions are connected through skills [20], [32]. We calculate the similarity among the questions

$$s_{(q_a, q_b)} = \text{sim}(\text{emb}(q_a), \text{emb}(q_b)) \quad (2)$$

where $s_{(q_a, q_b)}$ represents the similarity of question q_a and question q_b , $\text{emb}(\cdot)$ is a graph embedding algorithm (Node2vec is used here), and $\text{sim}(\cdot, \cdot)$ is a similarity calculation method (cosine similarity is used here). Then, we define a function $f : \mathcal{P}_c(X) \rightarrow \mathcal{E}^k$ to generate knowledge-aware hyperedges $\mathcal{E}^k$, where $\mathcal{P}_c(X)$ represents the set of all consecutive subsequences of historical responses X , further formulate f as

$$f(X) = \{[q_i, q_j] \mid q_i, q_j \in X, i < j, s_{(q_i, q_j)} > \gamma\} \quad (3)$$

where γ is the question similarity intervention parameter ensuring the effectiveness of the generated knowledge hyperedges. To mitigate the impact of hyperedge length for knowledge state, we calculate the weight of knowledge-aware hyperedge $e^k \in \mathcal{E}^k$ to form the incidence matrix

$$\mathbf{H}^k[n][e^k] = \begin{cases} |e^k| \times s_{(q_i, q_j)}, & q_i \in e^k, \quad i \leq n \leq j \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where $|\cdot|$ represents the length of the hyperedge e^k . Furthermore, we obtain the incidence matrix $\mathbf{H}^k$ of all knowledge-aware hyperedges at the adaptive scale.

Response intervals of varying scales exhibit distinct learning patterns. For instance, a sequence of consecutive incorrect answers within a certain interval may suggest that the learner is distracted in exercising, which is unable to reflect the true knowledge state. As shown in Fig. 3, to capture fine-grained learning patterns, we define a function $g : \mathcal{P}_c(X) \rightarrow \mathcal{E}^p$ to generate pattern-aware hyperedges $\mathcal{E}^p$, where $\mathcal{P}_c(X)$ represents the set of all consecutive subsequences of historical

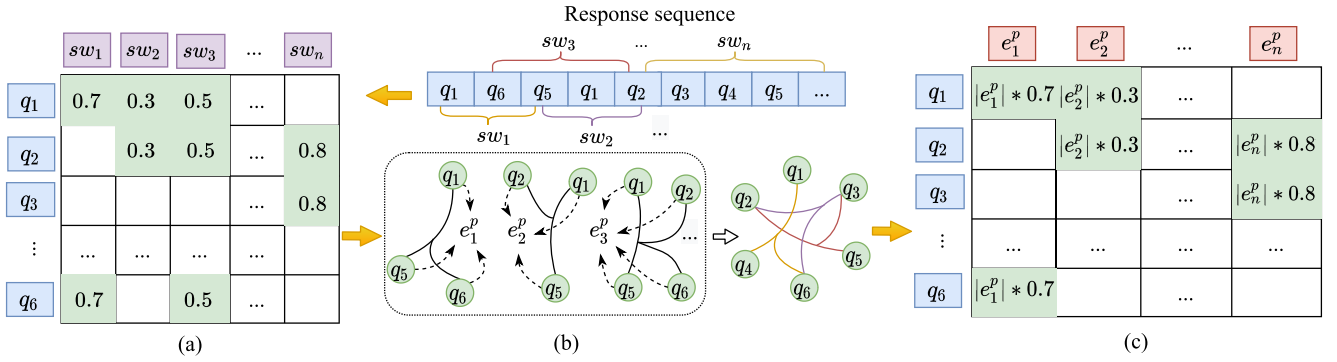


Fig. 3. Hyperedge distillation with pattern-aware. We first employ sliding windows to separate subsequences reflecting short-term learning patterns from the response sequence, and generate (a) pattern-aware matrix by calculating the correct answer rates of all subsequences; then (b) generate the pattern-aware hyperedges; and finally, (c) incidence matrix is generated based on the length of the hyperedges and the pattern-aware matrix.

responses X , further formulate g as

$$g(X) = \{[q_i, q_j] \mid q_i, q_j \in X, j - i = w\} \quad (5)$$

where w is the size of the sliding window employed to intercept response sequences. Similar to the generation process of knowledge-aware hyperedges, pattern-aware hyperedges also require hyperedge distillation because patterns reflect different learning states of learners. The learning pattern-aware control factor is calculated as

$$\rho_w(q_i, q_j) = \frac{1}{w} \sum_{k=i}^j a_k \quad (6)$$

where a_k is the correctness of the learner's response to the question q_k . We calculate the weight of pattern-aware hyperedge $e^p \in \mathcal{E}^p$ to form the incidence matrix

$$\mathbf{H}^p[n][e^p] = \begin{cases} |e^p| \times \rho_w(q_i, q_j), & q_n \in e^p, \quad i \leq n \leq j \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

Furthermore, we obtain the incidence matrix $\mathbf{H}^p$ of all pattern-aware hyperedges. By generating knowledge-aware hyperedges and pattern-aware hyperedges with adaptive scales, we derive the corresponding incidence matrixes, normalized as $\tilde{\mathbf{H}}^k$ and $\tilde{\mathbf{H}}^p$. Incidence matrixes serve as guides for the process of information propagation on multiview hypergraphs.

C. Dual-Channel Hypergraph Encoders

Traditional deep learning-based KT models rely on the single-channel encoded hidden vector to represent the learner's knowledge state [26], which often ignores the diversity of the knowledge state. The different views of response sequences reflect the learner's multigranularity knowledge states. To address this issue, we construct a global state hypergraph and a local state hypergraph from response sequences. Subsequently, we proposed dual-channel hypergraph encoders to capture learners' global and local knowledge states from the state hypergraphs.

Specifically, we first consider each learner's response sequence as a hyperedge and collectively form the global state hypergraph from all learners' interaction records. The simplified hypergraph convolution channel is proposed to model the global knowledge state from the global state hypergraph. Following the suggestions in [50], we remove the nonlinear

activation function and the learnable convolution filter matrix. Then, the spectral hypergraph convolution is defined as

$$\mathbf{X}_{\text{gs}}^{(l+1)} = \mathbf{D}_{\text{gs}}^{-1/2} \mathbf{H}_{\text{gs}}^{-1} \mathbf{B}_{\text{gs}} \mathbf{H}_{\text{gs}}^T \mathbf{D}_{\text{gs}}^{-1/2} \mathbf{X}_{\text{gs}}^{(l)} \quad (8)$$

where $\mathbf{X}_{\text{gs}}^{(l)} \in \mathbb{R}^{|\mathcal{V}| \times d^{(l)}}$ denotes the embedding matrix of the whole question set at l th layer, $\mathbf{D}_{\text{gs}} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ and $\mathbf{B}_{\text{gs}} \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{E}|}$ are diagonal matrices representing the degree of nodes and hyperedges, respectively. The hypergraph convolution operation follows the node-to-hyperedge and hyperedge-to-node information aggregation rules. To simplify matters, we calculate the question embeddings as $\mathbf{X}^{\text{global}} = (1/(L+1)) \sum_{l=0}^L \mathbf{X}_{\text{gs}}^{(l)}$.

There may be repeated questions in the response sequence, and learners provide different answers to these questions at different times. This renders the question embeddings learned by the simplified hypergraph convolution channel unsuitable for directly predicting learner performance. To address this issue, we concatenate the learner's answers to each moment's question. Moreover, the temporal ordering of the responses carries important information about the order in which the questions are attempted, which is necessary for accurately predicting the learner's knowledge state. Following the transformer [51], we integrate a learnable reverse position embedding with question embeddings. Specifically, the t th question embedding in the response sequence is given by

$$\bar{\mathbf{x}}_t^g = \tanh(\mathbf{W}_1[\mathbf{x}_t^g \oplus \mathbf{a}_t \oplus \mathbf{p}_{m-t+1}] + \mathbf{b}_1) \quad (9)$$

where $\mathbf{a}_t$ and $\mathbf{p}_{m-t+1}$ are learner's answer and reversed position embedding at moment t , $\mathbf{W}_1$ and $\mathbf{b}_1$ are the learnable parameters. Similar to [38] and [39], we calculate the question weights with a soft-attention mechanism as follows:

$$\varphi_{t,e} = \mathbf{f}_1^T \sigma(\mathbf{W}_2 \bar{\mathbf{x}}_e^g + \mathbf{W}_3 \bar{\mathbf{x}}_t^g + \mathbf{b}_2) \quad (10)$$

$$\boldsymbol{\theta}_g = \sum_{i=1}^m \varphi_{t,e} \bar{\mathbf{x}}_i \quad (11)$$

where $\bar{\mathbf{x}}_e^g = (1/m) \sum_{i=1}^m \mathbf{x}_m^g$ is the embedding of learner's response sequence with a length of m . $\mathbf{f}_1$, $\mathbf{W}_2$, $\mathbf{W}_3$, and $\mathbf{b}_2$ are attention parameters. $\boldsymbol{\theta}_g$ is the representation of global knowledge state.

As previously mentioned, the intervals of varying scales within the response sequence provide insight into the learner's local learning progress. To capture this local aspect, we construct local state hypergraphs for each learner via the proposed

adaptive scale hyperedge distillation module. Although the above convolution layer can propagate messages through hyperedges, it does not provide plenty of flexibility in adjusting for the importance of learning status for different nodes through different hyperedges. In other words, some responses connected by a hyperedge reveal the local knowledge state, but others may be interference. Therefore, we design a collaborative hypergraph convolution channel for local learning of hypergraph encoding.

Specifically, we assume that the responses connected by hyperedges can form hidden knowledge states, which indicates learners' local exercise orientation. Then, the representation for each local knowledge state is calculated by aggregating across its associated response nodes $\mathcal{N}_e$, denoted as $\mathbf{h}_e = \mathcal{AGG}(\{\mathbf{h}_i | i \in \mathcal{N}_e\})$, where $\mathcal{AGG}(\cdot)$ is an aggregation function (here is the mean aggregation) and $\mathbf{h}_i$ is the initialized vector for node i . We calculate attention score between response node i and its associative hyperedges Φ_i as follows:

$$\beta_{i,e} = \frac{\exp(\text{LeakyReLU}(\mathbf{f}_2^\top (\mathbf{h}_e \odot \mathbf{h}_i)))}{\sum_{i' \in \Phi_i} \exp(\text{LeakyReLU}(\mathbf{f}_2^\top (\mathbf{h}_e \odot \mathbf{h}_{i'})))} \quad (12)$$

where $\mathbf{f}_2$ is the attention vector and $\odot$ denotes the Hadamard product.

The adaptive scale hyperedge distillation module provides an incidence matrix $\mathbf{H}_{ls} = \{\mathbf{h}_{i,e}\} = \{\tilde{\mathbf{H}}^k \oplus \tilde{\mathbf{H}}^p\}$ of knowledge-aware hyperedges and pattern-aware hyperedges for local state hypergraph. Then, we collaborate with the weights of adaptive scales and attention scores to obtain an enhancement matrix $\tilde{\mathbf{H}} = \{\mathbf{h}_{i,e} \beta_{i,e}\}$. Following [52], we still consider two-order refinement of message passing via hypergraph convolution:

$$\mathbf{X}_{ls}^{(l+1)} = \mathbf{D}_{ls}^{-1/2} \tilde{\mathbf{H}}_{ls} \mathbf{B}_{ls}^{-1} \tilde{\mathbf{H}}_{ls}^T \mathbf{D}_{ls}^{-1/2} \mathbf{X}_{ls}^{(l)} \quad (13)$$

where $\mathbf{X}_{ls}^{(l+1)}$ denotes the questions embedding matrix in the local state hypergraph. Similar to the global state hypergraph encoding channel, the new question embedding concatenates the reverse position embedding and the learner's answer in the collaborative hypergraph convolution channel

$$\tilde{\mathbf{x}}_i^l = \sigma(\mathbf{W}_4 [\mathbf{x}_i^l \oplus \mathbf{a}_i \oplus \mathbf{p}_{m-t+1}] + \mathbf{b}_3) \quad (14)$$

where $\mathbf{W}_4$ and $\mathbf{b}_3$ are the learnable parameters. The local knowledge state is obtained by aggregating the question embeddings in the learner's response sequences, i.e., $\boldsymbol{\theta}_l = (1/n) \sum_{i=1}^n \tilde{\mathbf{x}}_i^l$.

The dual-channel hypergraph encoders module captures the high-order information of each question node, providing multigranularity knowledge states for learner performance prediction. We fuse the global and local knowledge states to capture the overall learning ability of the learner

$$\bar{\boldsymbol{\theta}} = \mathbf{f}_3^\top \sigma(\mathbf{W}_5 (\boldsymbol{\theta}_g + \boldsymbol{\theta}_l) + \mathbf{b}_4) \quad (15)$$

where $\mathbf{f}_3$, $\mathbf{W}_5$, $\mathbf{W}_5$, and $\mathbf{b}_4$ are the learnable parameters. We concatenate the final knowledge state $\bar{\boldsymbol{\theta}}$ and candidate question embedding $\mathbf{x}_{t+1}$, and project them to the output layer via a fully connected network with a sigmoid activation

$$\mathbf{y}_{t+1} = \sigma(\mathbf{W}_6 [\bar{\boldsymbol{\theta}} \oplus \mathbf{x}_{t+1}] + \mathbf{b}_5). \quad (16)$$

The objective function of predicting is the negative log-likelihood of the observed sequence of learners' exercising process from step 1 to $T + 1$. We also optimize a standard cross-entropy loss between the prediction $\mathbf{y}_{t+1}$ and the actual answer $\mathbf{a}_{t+1}$ as the objective function

$$\mathcal{L}_{KT} = - \sum_{t=1}^T (\mathbf{a}_{t+1} \log \mathbf{y}_{t+1} + (1 - \mathbf{a}_{t+1}) \log(1 - \mathbf{y}_{t+1})). \quad (17)$$

D. Cross-View Hypergraph Contrastive Learning

Due to the exploitation of high-order information between responses via hypergraph modeling, our model achieves significant performance (this is verified in subsequent experiments). However, the sparsity in response data may result in insufficiently rich structural features in the constructed state hypergraphs, thereby constraining the effectiveness of the dual-channel hypergraph encoders. Inspired by the self-supervised learning on simple graphs [49] and hypergraphs [38], we propose a cross-view hypergraph contrastive learning mechanism to enhance the representational capacity of multigranularity knowledge states encoded from the state hypergraphs.

Specifically, we transform the global and local state hypergraphs into line graphs. These graphs reflect the characteristics of response sequences in different views. Following [38], we define an incidence matrix $\mathbf{H}_{gg} \in \mathbb{R}^{M \times M}$ for the line graph of global state hypergraph $G(g)$, where M is the number of nodes. We initialize the node embedding of $G(g)$

$$\boldsymbol{\psi}_i^{(0)} = \frac{1}{|e_i^{gs}|} \sum_{j \in e_i^{gs}} \mathbf{x}_j^{(0)} \quad (18)$$

where $\boldsymbol{\psi} \in \Psi$, and e_i^{gs} is a hyperedge of global state hypergraph. Then the graph convolution process is defined as

$$\Psi_{gg}^{(l+1)} = \mathbf{D}_{gg}^{-1} \mathbf{H}_{gg} \Psi_{gg}^{(l)} \quad (19)$$

where $\Psi_{gg}^{(l)}$ is the node embedding in $G(g)$ at l th layer. $\mathbf{D}_{gg} \in \mathbb{R}^{M \times M}$ is a degree matrix. After l -layer graph convolution, we get the final node embedding $\Psi_{gg}' = (1/(L+1)) \sum_{l=0}^L \Psi_{gg}^{(l)}$, which also represents the embedding of learner response sequence.

Similarly, we transform the local state hypergraph into a line graph. After l -layer convolution, we can obtain the embedding of nodes $\Psi_{lg} = (1/(L+1)) \sum_{l=0}^L \Psi_{lg}^{(l)}$, representing the embedding of hyperedges in local state hypergraphs. The learner's local knowledge state can be obtained by aggregating related hyperedges $\Psi_{lg}' = (1/N) \sum_{n=1}^N \Psi_{lg}^n$, where N is the number of hyperedges in one learner response sequence.

So far, we have obtained the embedding of the response sequence reflecting the learner's knowledge state via multiple view encoders. Mutual information between knowledge states of local views can be maximized by contrastive learning between the hypergraph and the corresponding line graph. The mapping relationship between these two knowledge state embeddings can be used to enhance the supervisory signal. In other words, the embeddings of the same learner in the hypergraph and the corresponding line graph are treated as

ground-truth pairs, otherwise they are treated as negative pairs. Following the suggestions in [38], we employ mutual information noise contrastive estimation (InfoNCE) [47] as our contrastive learning objective:

$$\mathcal{L}_{\text{gs}} = -\log \sigma \left(s \left(\theta_i^{\text{gs}}, \psi_i^{\text{gs}} \right) \right) - \log \sigma \left(1 - s \left(\tilde{\theta}_i^{\text{gs}}, \psi_i^{\text{gs}} \right) \right) \quad (20)$$

$$\mathcal{L}_{\text{ls}} = -\log \sigma \left(s \left(\theta_i^{\text{ls}}, \psi_i^{\text{ls}} \right) \right) - \log \sigma \left(1 - s \left(\tilde{\theta}_i^{\text{ls}}, \psi_i^{\text{ls}} \right) \right) \quad (21)$$

where $\theta_i^{\text{gs}} \in \Theta^{\text{gs}}$ and $\theta_i^{\text{ls}} \in \Theta^{\text{ls}}$ are encoded from the proposed dual-channel hypergraph encoders to represent global and local knowledge state for student i (more details in Section IV-C), $\tilde{\theta}_i^*$ is the negative sample obtained by corrupting Θ^* with row-wise and column-wise shuffling, $\mathcal{L}_{\text{gs}}$ and $\mathcal{L}_{\text{ls}}$ denote the global cross-view contrastive loss and the local cross-view contrastive loss, and $s(\cdot, \cdot)$ is the discriminator function implemented by the dot product.

The proposed dual-channel hypergraph encoders capture a distinctive aspect of the learner's knowledge state. Each encoder has to work cooperatively to maximize the overall modeling capacity. Therefore, we introduce global-local cross-view contrastive learning, which is calculated as

$$\mathcal{L}_{\text{gl}} = -\log \sigma \left(s \left(\theta_i^{\text{ls}}, \theta_i^{\text{gs}} \right) \right) - \log \sigma \left(1 - s \left(\tilde{\theta}_i^{\text{ls}}, \theta_i^{\text{gs}} \right) \right). \quad (22)$$

Finally, we unify the KT task with the auxiliary self-supervised task. All the losses are combined to form a joint optimization framework as follows:

$$\mathcal{L} = \min \mathcal{L}_{\text{KT}} + \lambda (\mathcal{L}_{\text{ls}} + \mathcal{L}_{\text{gs}} + \mathcal{L}_{\text{gl}}) \quad (23)$$

where λ is a coefficient to control the trade-off between KT task constraints and contrastive learning task constraints. More details of settings are specified in the subsequent experiments.

E. Complexity Analysis

1) *Time Complexity*: The time complexity of the channel encoding global knowledge state through a simplified hypergraph convolution network is $O(N_g E_g d)$, where N_g and E_g are the number of nodes and hyperedges in the global state hypergraph, respectively. For another channel encoding the local knowledge state, the computational cost includes hyperedge distillation and collaborative hypergraph convolution network. The time complexity of the distillation process of knowledge-aware hyperedge and pattern-aware hyperedge is $O(N_l + w N_l)$, where N_l is the number of nodes in the local state hypergraph. The collaborative hypergraph convolution network takes $O(2N_l E_l d)$ time complexity, where E_l is the number of hyperedges in the local state hypergraph. The time complexity of the line graph convolution network in the cross-view hypergraph contrastive learning module is $O(E_g d + E_l d)$. Since $N_g, N_l, E_g, E_l \gg d$, $N_g = N_l$ denote the number of questions, the overall time complexity of HyperKT therefore is $O((E_g + 2E_l)Nd)$.

We compare our HyperKT model with several state-of-the-art methods including GIKT [19], GFLDKT [20], collaborative knowledge tracing (CoKT) [15], and learning process-consistent knowledge tracing to distinct students' progress rates (LPKT-S) [14] (details of introduction and performance for these methods are in Section V-C). GIKT is a graph-based model with time complexities of

TABLE I
DATASET STATISTICS

Statistics	Assist2009	EdNet1	EdNet2
# of learners	3841	5000	5000
# of questions	15911	11718	11427
# of skills	123	189	188
# of records	258896	498374	394986
Avg. questions per skill	156.1	136.8	133.7
Avg. skills per question	1.2	2.2	2.2

$O((Q + S)d^2 + 2T(Q + S)d)$, where Q is the number of questions, S is the number of skills, T is the length of response sequences, and $Q \gg S$. GFLDKT is also a graph-based method, and its computational cost is close to that of GIKT. CoKT takes $O(LT + KTd)$ time complexity, where $L = E_g$ denote the number of learners and K is the number of similar sequences retrieved. The time complexity of LPKT-S is $O(T(Q + S)d)$, which mainly depends on the operation of long short-term memory (LSTM). Hyperedge distillation makes the number of hyperedges generated in the sequence usually less than the length of the sequence and $T \gg L, Q$. Therefore, the time efficiency of HyperKT is better than GIKT and slightly worse than CoKT and LPKT-S.

2) *Space Complexity*: For space complexity, the memory consumption of HyperKT mainly derives from dual-channel hypergraph encoders. For the local knowledge state encoder channel, the space complexity of the question similarity matrix is $O(N^2)$, the size of the adjacency matrix of the local state hypergraph is NE_l , and the size of the node embedding matrix is Nd , where N is the number of questions. The global knowledge state encoder channel takes $O(N^2 + Nd)$ space complexity. In addition, two line graph convolution networks in the cross-view contrastive learning module take $O(L_{\text{inp}}^2 + L_{\text{inp}}d)$ space complexity, where L_{inp} is the number of learners input to the HyperKT at a time. Therefore, HyperKT takes $O(2N^2 + NE_l + 2Nd + L_{\text{inp}}^2 + L_{\text{inp}}d)$ space complexity.

GIKT takes $O(T_{\text{inp}} \bar{N}_{\text{nei}} d + (Q + S)d + QS)$ space complexity with graph convolutional network (GCN) and the interaction layer, where T_{inp} is the number of response records input to the GIKT at a time and $\bar{N}_{\text{nei}} = 4$ is the average number of neighbors of the question node. The space complexity of CoKT is $O(T_{\text{inp}} d + 3|R|d)$ includes memory consumption of similar sequence retrieval and the KT task, where $|R|$ is the number of similar response sequences. LPKT-S takes $O((T_{\text{inp}} + Q + 2S)d + QS)$ space complexity. Since graphs are usually sparse, $T_{\text{inp}} d > N^2, NE_l$, and $N = Q \gg L_{\text{inp}}, S$. HyperKT exhibits a more efficient space complexity relative to GIKT, approaching that of CoKT and LPKT-S.

In summary, our model achieves SOTA performance on the KT task without sacrificing computational overhead.

V. EXPERIMENTS

In this section, a series of experiments on several real-world datasets are detailed to evaluate the performance of HyperKT by predicting future learner responses.

A. Datasets

We use three real-world public datasets to evaluate the effectiveness of the proposed model. The statistics of all

datasets are shown in Table I. The brief descriptions of the datasets are as follows.

- 1) *ASSISTments 2009 (Assist2009)*¹: ASSISTments is an online system for teaching and evaluating learner mathematics. This dataset is publicly available. Following [32], we filter records without knowledge concepts as well as scaffolding questions.
- 2) *EdNet-KT1 (EdNet1)*²: EdNet is a dataset of all learner systems interactions collected over two years by Santa, which is a multiplatform AI tutoring service. Following the suggestions of [14], we use the EdNet-KT1 dataset consisting of learners' question-solving logs and randomly extract 5000 learners' records.
- 3) *EdNet-KT2 (EdNet2)*²: EdNet-KT2 consists of the actions related to question-solving activities, such as the study mode or the intermediate responses provided by the learner. Similar to the process of EdNet1, we randomly extract 5000 learners' records from the EdNet-KT2 dataset, which retains the relevant data of the learner's responses.

B. Implementation Details

Following [16], to ensure fairness and the model's universality, we fix the length of each learner's response sequence to 50 in all datasets. For sequences shorter than the fixed length, we use zero vectors to pad them. For all datasets, we split 80% data for training and validation, and 20% for testing. For all comparison methods and variants of HyperKT, we repeat each experiment ten times and report the average of the results with standard deviations. To compare the performance of the proposed model and selected baselines, we employ widely used evaluation metrics in experiments [6], [21], the area under the curve (AUC), accuracy (ACC), mean absolute error (MAE), and root-mean-square error (RMSE).

In our implementation of HyperKT, the dimension size of question embeddings, multilevel knowledge states, and reversed position embeddings are set to 100. In the adaptive scale hyperedge distillation module, the question similarity intervention parameter γ is set to 0.7 (a small γ introduces interference to knowledge-aware hyperedges, while a large γ results in sparsity in knowledge-aware hyperedges.), and the size of the sliding window w for pattern-aware control is set to [3, 4, 5]. The number of convolution layers of all views is unified to 3. In the cross-view hypergraph contrastive learning module, the trade-off coefficient λ for the KT task and contrastive learning task is set to 0.002. To optimize HyperKT, the learning rate and dropout rate are set as 0.001 and 0.6. We perform a sensitivity analysis on the setting of key hyperparameters. The hyperparameters for all baselines are carefully tuned to their best performances.

C. Baseline Methods

Table II shows the summary of various characteristics involved in all baseline models and HyperKT. The details of all baselines are as follows.

TABLE II
CHARACTERISTICS OF ALL MODELS. THE CHARACTERISTICS OF THE MODELS INCLUDE RNNs-BASED, FACTORIZATION MACHINE-BASED (FM), MODELING LEARNING BEHAVIOR-BASED (LB), ATTENTION MECHANISM-BASED, CONTRASTIVE LEARNING-BASED (CL), AND GRAPH-BASED. ✓* REPRESENTS THE HYPERGRAPH-BASED MODEL

Model	Characteristics					
	RNNs	FM	LB	Attention	CL	Graph
DKT	✓					
KTM		✓				
DKTForgetting	✓		✓			
SAKT				✓		
AKT			✓	✓		
GIKT	✓		✓			✓
HawkesKT			✓			
IEKT	✓		✓			
CoKT	✓		✓	✓		
GFLDKT	✓		✓			✓
LPKT-S	✓		✓			
HEMR					✓	✓*
CentSmoothie					✓	✓*
AdaS				✓	✓	✓*
HyperKT				✓	✓	✓*

- 1) **DKT** [11] is the first method that introduces deep learning to model KT tasks. We utilize LSTM to implement DKT.
- 2) **KTM** [8] utilizes factorization machines to model response sequences between features which include question id, skill id, historical responses on different skills, and temporal.
- 3) **DKTForgetting** [13] is a DKT-based model that treats past trials and time gaps as forgetting-related informative features.
- 4) **Self Attentive Knowledge Tracing (SAKT)** [30] is a deep-learning method based on self-attention mechanism.
- 5) **GIKT** [19] obtains the high-order embeddings of the questions from the question-skill graph, and introduces the recap module to simulate the learning process.
- 6) **Attentive Knowledge Tracing (AKT)** [9] is also an attention-based approach that has improvements in attention mechanism with forgetting feature, and the representations of questions and skills.
- 7) **HawkesKT** [16] works on modeling time-sensitive effects in KT task. HawkesKT proves the existence of temporal cross-effects in the learning process through experiments and introduces Hawkes to model these effects.
- 8) **Individual Estimation Knowledge Tracing (IEKT)** [31] introduces the cognition level estimation module and knowledge acquisition sensitivity estimation module to the KT task.
- 9) **CoKT** [15] obtains inter-learner information by retrieving sequences of peer learners who have comparable question-answering experiences, then combines it with intra-learner information to trace knowledge states.
- 10) **GFLDKT** [20] designs two gating-controlled mechanisms to model explicitly forgetting and learning behaviors. In addition, it is a graph-based method, which uses the Q-matrix-induced partial-order-based method to pre-train the exercise embedding.

¹<https://sites.google.com/site/assistmentsdata/home/2009-2010-assistment-data>

²<https://github.com/riiid/ednet>

TABLE III

PERFORMANCES OF ALL COMPARISON METHODS ON THREE DATASETS. THE PERFORMANCE OF OUR METHOD AND THE BEST-PERFORMED BASELINE ARE PRESENTED IN BOLD AND UNDERLINED, RESPECTIVELY. THE VALUE AFTER $\pm$ IS THE STANDARD DEVIATION

Model	Assist2009				EdNet1			
	AUC	ACC	MAE	RMSE	AUC	ACC	MAE	RMSE
DKT	74.84 (± 0.74)	71.70 (± 0.68)	36.85 (± 0.33)	43.66 (± 0.21)	68.77 (± 0.52)	63.90 (± 0.44)	44.40 (± 0.25)	47.00 (± 0.16)
KTM	73.21 (± 1.84)	68.78 (± 1.63)	39.04 (± 2.25)	44.63 (± 1.81)	72.37 (± 1.46)	67.16 (± 1.02)	40.52 (± 0.78)	46.08 (± 0.43)
DKTForgetting	74.68 (± 0.63)	70.75 (± 0.57)	37.46 (± 0.59)	43.93 (± 0.33)	68.15 (± 0.64)	63.49 (± 0.44)	42.72 (± 0.21)	47.47 (± 0.29)
SAKT	69.83 (± 1.12)	68.52 (± 1.39)	40.78 (± 0.77)	45.70 (± 0.63)	66.07 (± 1.46)	62.10 (± 1.24)	44.89 (± 1.06)	47.81 (± 0.83)
AKT	72.10 (± 1.20)	68.97 (± 0.87)	38.32 (± 0.69)	45.68 (± 1.07)	71.90 (± 1.22)	66.78 (± 0.73)	39.71 (± 0.56)	46.70 (± 0.74)
GIKT	79.48 (± 0.53)	74.24 (± 0.67)	32.37 (± 0.42)	42.54 (± 0.25)	74.20 (± 0.81)	68.18 (± 0.42)	39.80 (± 0.44)	45.63 (± 0.39)
HawkesKT	73.46 (± 1.62)	69.42 (± 0.81)	38.26 (± 0.98)	44.82 (± 0.75)	71.66 (± 1.17)	66.74 (± 0.93)	41.74 (± 0.62)	46.42 (± 0.37)
IEKT	77.27 (± 1.07)	72.60 (± 1.33)	34.94 (± 0.78)	43.06 (± 0.59)	73.19 (± 1.21)	67.54 (± 1.02)	40.94 (± 0.88)	45.66 (± 0.26)
CoKT	77.03 (± 0.60)	72.23 (± 0.34)	37.10 (± 0.19)	42.96 (± 0.26)	72.44 (± 0.51)	66.77 (± 0.38)	42.29 (± 0.26)	45.84 (± 0.13)
GFLDKT	73.78 (± 0.83)	68.01 (± 0.51)	38.09 (± 0.37)	44.97 (± 0.24)	73.42 (± 0.66)	67.43 (± 0.33)	41.35 (± 0.09)	45.64 (± 0.28)
LPKT-S	78.32 (± 0.62)	73.19 (± 0.55)	34.26 (± 0.33)	43.39 (± 0.39)	75.18 (± 0.51)	68.77 (± 0.48)	36.95 (± 0.17)	45.42 (± 0.22)
HEMR	75.16 (± 0.52)	71.09 (± 0.33)	36.95 (± 0.42)	46.07 (± 0.31)	71.81 (± 0.46)	64.24 (± 0.31)	43.15 (± 0.14)	47.83 (± 0.27)
CentSmoothie	77.22 (± 0.61)	72.29 (± 0.56)	34.35 (± 0.49)	43.97 (± 0.44)	73.16 (± 0.38)	66.31 (± 0.46)	40.75 (± 0.34)	45.93 (± 0.31)
AdaS	76.36 (± 0.31)	71.87 (± 0.26)	36.49 (± 0.23)	46.17 (± 0.14)	72.51 (± 0.24)	65.54 (± 0.27)	41.18 (± 0.16)	46.26 (± 0.11)
HyperKT	81.74 (± 0.36)	77.34 (± 0.29)	30.62 (± 0.31)	41.13 (± 0.23)	77.27 (± 0.16)	70.97 (± 0.25)	34.47 (± 0.17)	44.39 (± 0.22)

Model	EdNet2			
	AUC	ACC	MAE	RMSE
DKT	61.40 (± 0.44)	58.24 (± 0.37)	47.46 (± 0.20)	48.95 (± 0.13)
KTM	65.09 (± 1.21)	60.82 (± 1.04)	46.29 (± 0.47)	48.34 (± 0.28)
DKTForgetting	61.83 (± 0.43)	58.64 (± 0.36)	46.78 (± 0.19)	49.17 (± 0.14)
SAKT	56.77 (± 1.23)	51.29 (± 1.36)	49.41 (± 0.92)	50.00 (± 0.69)
AKT	63.27 (± 1.06)	59.21 (± 0.91)	46.13 (± 0.47)	48.92 (± 0.68)
GIKT	65.12 (± 0.44)	<u>61.53</u> (± 0.36)	45.56 (± 0.21)	48.68 (± 0.35)
HawkesKT	64.02 (± 1.17)	60.07 (± 1.32)	46.85 (± 0.66)	48.55 (± 0.40)
IEKT	65.30 (± 0.78)	61.43 (± 0.59)	45.39 (± 0.33)	49.02 (± 0.42)
CoKT	63.87 (± 0.55)	60.42 (± 0.61)	45.90 (± 0.39)	48.57 (± 0.24)
GFLDKT	<u>65.56</u> (± 0.74)	61.22 (± 0.38)	46.70 (± 0.16)	<u>48.15</u> (± 0.17)
LPKT-S	65.30 (± 0.71)	61.35 (± 0.66)	45.83 (± 0.29)	48.56 (± 0.38)
HEMR	63.49 (± 0.55)	60.18 (± 0.42)	46.38 (± 0.39)	48.82 (± 0.26)
CentSmoothie	64.53 (± 0.37)	61.36 (± 0.37)	45.82 (± 0.24)	48.65 (± 0.19)
AdaS	63.92 (± 0.21)	60.27 (± 0.19)	46.45 (± 0.14)	48.73 (± 0.13)
HyperKT	68.68 (± 0.34)	63.75 (± 0.27)	44.02 (± 0.15)	47.22 (± 0.21)

- 11) **LPKT-S** [14] considers the learning gain, forgetting, and personal progress rate characteristics during the learning process to improve the performance of KT.
- 12) **HEMR** [23] introduces an efficient hypergraph embedding method based on Random Walks and Word2Vec for music recommendation. We remove the local state hypergraph encoding channel in HyperKT and replace the global state hypergraph encoder with the hypergraph embedding method in HEMR for the KT task.
- 13) **CentSmoothie** [35] introduces a hypergraph neural network based on the central-smoothing strategy for predicting drug–drug interactions. Similar to HEMR, we replace the global state hypergraph encoder with the hypergraph neural network in CentSmoothie.
- 14) **AdaS** [47] devises a new adaptive sampling strategy to improve the performance of graph contrastive learning. We replace the augmented views in HyperKT with the hypergraphs adaptively sampled from the local and global state hypergraphs.

D. Overall Performance

The performances of all baselines and our HyperKT model are shown in Table III. We find the following observations. The proposed HyperKT model achieves the best performance compared to the baselines on three datasets. We attribute such improvements to the following aspects: 1) the adaptive scale hyperedge distillation module helps HyperKT extract

non-pairwise high-order features from learner response sequences; 2) the dual-channel hypergraph encoders module provides message propagation mechanisms for different hypergraph views, which is applied to discover learners' multi-granularity knowledge states; and 3) the cross-view hypergraph contrastive learning module enhances the supervisory signal in the hypergraph, which is considered an auxiliary task to improve the prediction performance.

In addition, the baselines KTM and HawkesKT based on temporal effect modeling perform better than DKT, which proves that temporal features can effectively improve model performance. The attention-based baseline AKT achieves significant improvements on all datasets compared to DKT, which confirms that the attention mechanism introduced into the KT task is working. Modeling various learning behaviors during the learning process (such as forgetting and learning gains) can help improve performance prediction, which can be reflected by comparing the performance differences between some baselines (KTFForgetting, IEKT, CoKT, GFLDKT, and LPKT-S) and DKT. Notably, the graph-based KT methods GIKT and GFLDKT show excellent results in most cases, which proves that the effectiveness of combining question modeling and learning behavior modeling is excellent. Even though the approaches described above have made significant advances to the KT task, they are still limited by pairwise relations. Benefiting from the ability of hypergraphs to capture higher-order relationships among responses, HyperKT achieves SOTA performance. HyperKT outperforms HEMR and CentSmoothie,

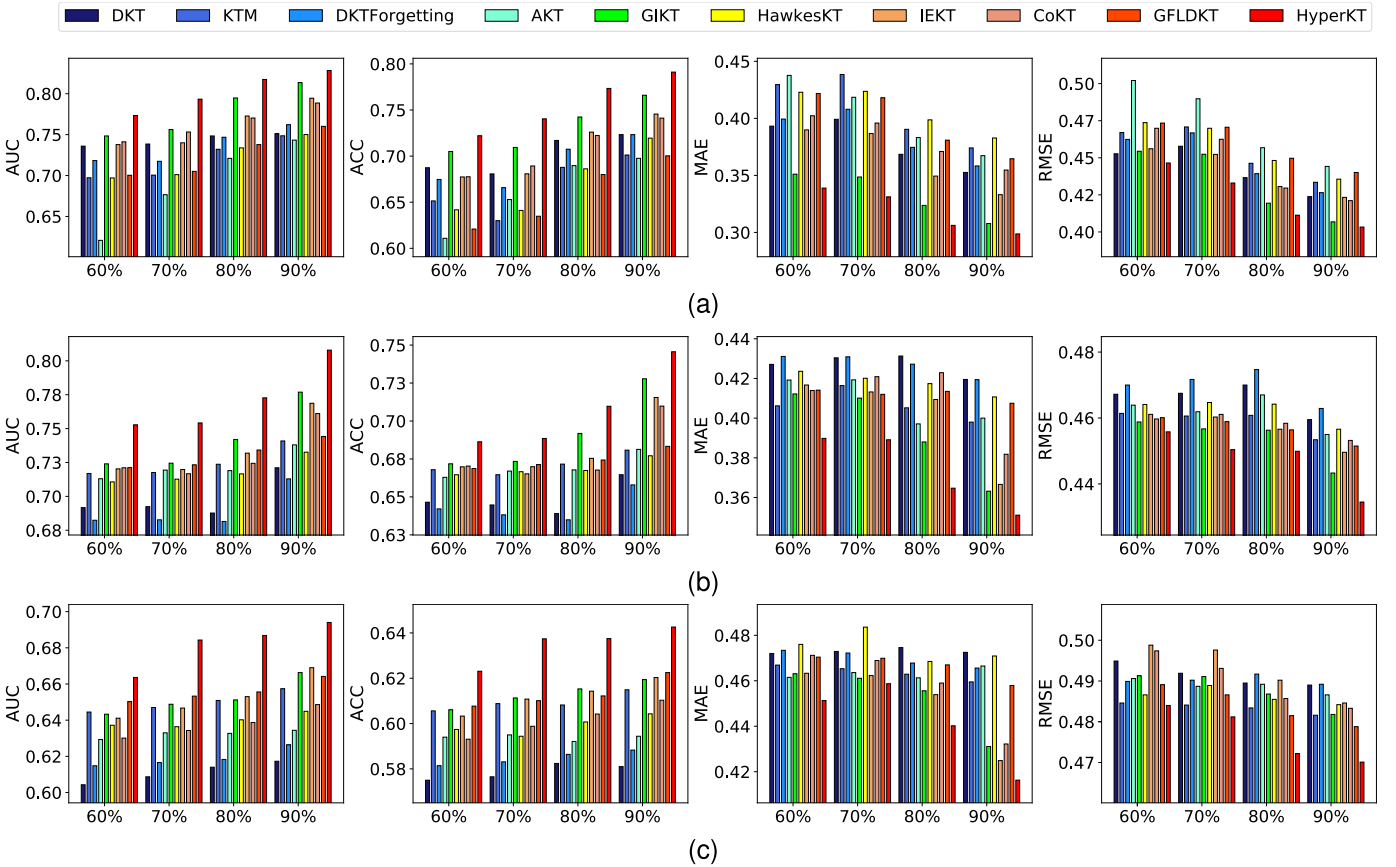


Fig. 4. Comparison of learner performance prediction results with different train-test split ratios on all datasets. (a) Assist2009. (b) EdNet1. (c) EdNet2.

illustrating the superior capability of dual-channel hypergraph encoders in capturing multigranularity knowledge states. AdaS, constrained by the randomness of sampling, may lose some crucial information, while HyperKT demonstrates a notable advantage by transforming state hypergraphs into line graphs as the augment views.

Finally, we compare the performance of the baseline model and HyperKT under different train-test ratio dataset splits. The prediction results on the three datasets are shown in Fig. 4, and it is clear that HyperKT outperforms the baselines in almost all cases. Even with a small training set ratio, HyperKT still shows excellent performance, which verifies the generalization ability of the HyperKT model.

E. Ablation Study

To evaluate the effectiveness of several key modules in HyperKT, we performed extensive ablation studies as shown in Table IV. Analyzing such performance comparison, we have the following observations.

1) *Hyperedge Generation*: We compare the performance without similarity among questions (-Sim), without knowledge-aware hyperedges (-KA), without pattern-aware hyperedges (-PA). The results suggest that deleting any type of hyperedge would degrade performance. In addition, without calculating the similarity among questions for the hyperedge distillation module means that the start and end of the knowledge-aware interval are the same questions. This varia-

tion results in the sparsity of the knowledge-aware hyperedge, which leads to a degradation in model performance.

2) *Hypergraph Views*: We compare the performance without local state hypergraph (-HyperLS), without global state hypergraph (-HyperGS). The results for these two variants of HyperKT demonstrate that the single-channel hypergraph encoder significantly degrades the model's performance. The explanation for these results is that the proposed dual-channel hypergraph encoders learn historical response sequences along two channels: the learner's long-term learning process and the short-term learning process. These two processes reflect the learner's mastery states of global knowledge and local knowledge. In addition, -HyperGS discards the local state hypergraph constructed through hyperedge distillation, making it difficult to capture complex high-order features among responses, thus showing the worst performance.

3) *Hypergraph Encoders*: We compare the performance without hypergraph attention (-Att), without hypergraph convolution (-Conv), without hyperedge distillation weights matrix H (-H). Removing hypergraph attention cannot capture the adaptive influence between hyperedges and nodes, and removing hypergraph convolution weakens the expressiveness and generalization ability of the model, both of the two variants affect the performance of HyperKT. Additionally, the knowledge flows or learning patterns inherent in hyperedges with adaptive scales are distinct. Removing the hyperedge distillation weight matrix implies that the knowledge-aware hyperedge and pattern-aware hyperedge have the same initial

TABLE IV
ABLATION STUDY ON KEY COMPONENTS OF HYPERKT. THE VALUE AFTER $\pm$ IS THE STANDARD DEVIATION

Model		Assist2009				EdNet1			
		AUC	ACC	MAE	RMSE	AUC	ACC	MAE	RMSE
Hyperedge Generation	-Sim	81.12 (± 0.33)	77.01 (± 0.25)	31.14 (± 0.18)	41.68 (± 0.24)	76.99 (± 0.41)	70.08 (± 0.52)	35.12 (± 0.26)	45.13 (± 0.31)
	-KA	80.94 (± 0.25)	76.98 (± 0.33)	31.28 (± 0.14)	41.83 (± 0.27)	76.32 (± 0.31)	69.76 (± 0.19)	35.46 (± 0.09)	45.46 (± 0.15)
	-PA	81.06 (± 0.26)	77.16 (± 0.31)	31.19 (± 0.28)	41.77 (± 0.21)	77.01 (± 0.29)	69.83 (± 0.33)	35.02 (± 0.18)	45.05 (± 0.19)
Hypergraph Views	-HyperLS	78.23 (± 0.65)	75.16 (± 0.57)	33.08 (± 0.38)	43.01 (± 0.25)	74.66 (± 0.46)	67.45 (± 0.61)	36.69 (± 0.42)	45.86 (± 0.27)
	-HyperGS	78.44 (± 0.59)	75.52 (± 0.48)	32.57 (± 0.32)	42.55 (± 0.27)	74.89 (± 0.61)	67.29 (± 0.55)	36.83 (± 0.39)	46.01 (± 0.26)
Hypergraph Encoders	-Att	80.33 (± 0.21)	76.34 (± 0.34)	31.32 (± 0.19)	41.88 (± 0.11)	76.42 (± 0.23)	70.11 (± 0.25)	35.66 (± 0.17)	45.29 (± 0.22)
	-Conv	79.67 (± 0.36)	76.15 (± 0.31)	31.99 (± 0.27)	42.33 (± 0.15)	76.10 (± 0.41)	69.32 (± 0.38)	35.78 (± 0.22)	45.43 (± 0.17)
	-H	80.85 (± 0.35)	76.81 (± 0.33)	31.41 (± 0.24)	42.02 (± 0.16)	76.11 (± 0.38)	69.53 (± 0.29)	35.65 (± 0.21)	45.20 (± 0.11)
Cross-view CL	$-\mathcal{L}_{ls}$	79.16 (± 0.45)	76.34 (± 0.16)	31.06 (± 0.33)	41.57 (± 0.28)	76.52 (± 0.35)	69.55 (± 0.42)	35.16 (± 0.12)	45.06 (± 0.27)
	$-\mathcal{L}_{gs}$	80.02 (± 0.29)	76.70 (± 0.22)	30.07 (± 0.15)	41.33 (± 0.21)	76.75 (± 0.25)	69.78 (± 0.31)	35.25 (± 0.11)	45.18 (± 0.08)
	$-\mathcal{L}_{gl}$	80.45 (± 0.32)	77.08 (± 0.27)	31.35 (± 0.16)	41.80 (± 0.13)	76.48 (± 0.20)	70.22 (± 0.24)	34.28 (± 0.09)	44.61 (± 0.14)
HyperKT		81.74 (± 0.36)	77.34 (± 0.29)	30.62 (± 0.31)	41.13 (± 0.23)	77.27 (± 0.16)	70.97 (± 0.25)	34.47 (± 0.17)	44.39 (± 0.22)

Model		EdNet2			
		AUC	ACC	MAE	RMSE
Hyperedge Generation	-Sim	68.08 (± 0.23)	63.35 (± 0.34)	44.18 (± 0.17)	47.66 (± 0.15)
	-KA	67.99 (± 0.42)	63.10 (± 0.22)	44.43 (± 0.21)	47.81 (± 0.23)
	-PA	68.12 (± 0.47)	63.44 (± 0.29)	44.26 (± 0.14)	47.72 (± 0.21)
Hypergraph Views	-HyperLS	65.17 (± 0.48)	60.77 (± 0.46)	46.84 (± 0.33)	49.03 (± 0.27)
	-HyperGS	66.33 (± 0.45)	61.24 (± 0.39)	46.70 (± 0.24)	48.86 (± 0.28)
Hypergraph Encoder	-Att	67.85 (± 0.32)	62.35 (± 0.21)	45.77 (± 0.19)	48.52 (± 0.16)
	-Conv	67.51 (± 0.38)	62.21 (± 0.37)	45.81 (± 0.14)	48.77 (± 0.22)
	-H	67.33 (± 0.47)	63.03 (± 0.24)	44.56 (± 0.21)	48.08 (± 0.14)
Cross-view CL	$-\mathcal{L}_{ls}$	68.04 (± 0.33)	62.84 (± 0.38)	45.12 (± 0.26)	47.99 (± 0.19)
	$-\mathcal{L}_{gs}$	68.12 (± 0.33)	62.65 (± 0.29)	45.09 (± 0.17)	47.77 (± 0.21)
	$-\mathcal{L}_{gl}$	68.27 (± 0.25)	63.08 (± 0.27)	44.83 (± 0.16)	48.03 (± 0.12)
HyperKT		68.68 (± 0.34)	63.75 (± 0.27)	44.02 (± 0.15)	47.22 (± 0.21)

weight, which causes HyperKT to fail to accurately model the learners' knowledge state.

4) *Cross-View CL*: We compare the performance without cross-view contrastive loss between local state hypergraph and corresponding transformed line graph ($-\mathcal{L}_{ls}$), without cross-view contrastive loss between global state hypergraph and corresponding transformed line graph ($-\mathcal{L}_{gs}$), without cross-view contrastive loss between local and global state hypergraphs ($-\mathcal{L}_{gl}$). The performance drop of variants $-\mathcal{L}_{ls}$ and $-\mathcal{L}_{gs}$ is significant, which demonstrates the superiority of learning discriminative information between state hypergraphs and corresponding line graphs. The result of $-\mathcal{L}_{gl}$ verifies that maximizing mutual information between multilevel knowledge states is beneficial to improve model performance.

F. Parameter Sensitivity Analysis

To study the impacts of dual-channel hypergraph convolution networks' depth, we range the numbers of layers in the networks within $\{1, 2, 3, 4, 5\}$ (note that we set the same number of convolution layers for all view encoders). According to the results presented in Fig. 5, HyperKT is not very sensitive to the number of layers on EdNet1 and EdNet2, a three-layer setting is the best. However, on Assist09, a two-layer network achieves the best performance. The reasons for this phenomenon may be twofold. One is that the sparse correlation between questions and skills in Assist2009 leads

to inaccurate similarity calculation between questions, which introduces noise to the knowledge hyperedges. Another reason is that the length of the response sequence in the Assist2009 dataset is shorter, which results in insufficient pattern-aware hyperedges. These reasons make it unnecessary to have too many layers of hypergraph convolution to ensure the generalization ability of HyperKT on the Assist2009 dataset. Besides, with the number of layers increasing, the performance on four metrics drops for all datasets. The possible cause could be the increasingly over-smoothed for learned representations of questions and knowledge states.

We introduce a hyperparameter λ to HyperKT to control the effect of the self-supervised learning task. To investigate the influence of the self-supervised learning task based on cross-view hypergraph contrastive learning, we report the performance of HyperKT with a set of representative λ values $\{2 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}, 3 \times 10^{-3}, 4 \times 10^{-3}\}$. According to the results presented in Fig. 6, the KT task achieves decent gains when jointly optimized with the self-supervised task. For all datasets, learning with $\lambda = 2 \times 10^{-3}$ can boost all metrics, and with the increase of λ , the performance declines. This phenomenon may be caused by gradient conflicts between KT tasks and self-supervised learning tasks. A too-small self-supervised learning task control factor degrades the performance of the model, which proves that cross-view hypergraph contrastive learning is beneficial to model learners' knowledge states.

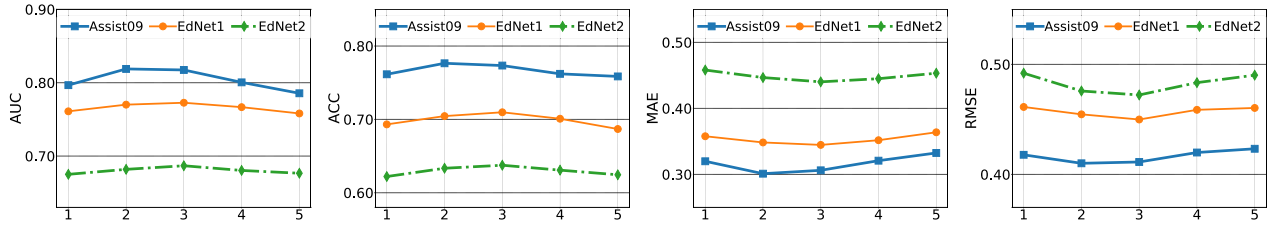


Fig. 5. Impacts of the size for layer.

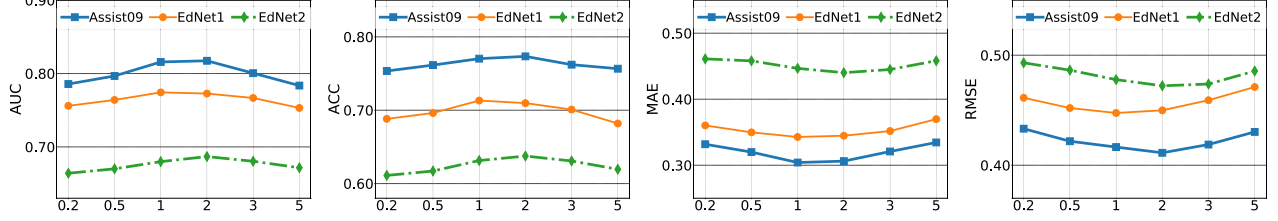
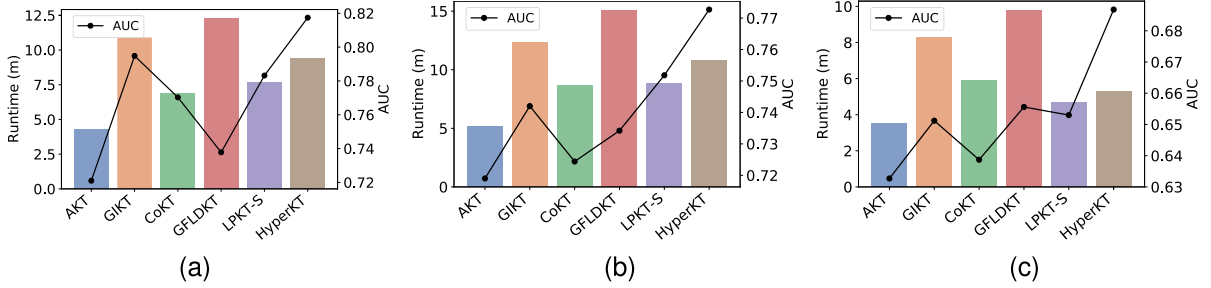
Fig. 6. Impact of the constraint λ of self-supervised learning. The value of the horizontal axis is $\lambda \times 10^3$.

Fig. 7. Comparison of runtime consumption (minutes) and AUC on all datasets. (a) Assist2009. (b) EdNet1. (c) EdNet2.

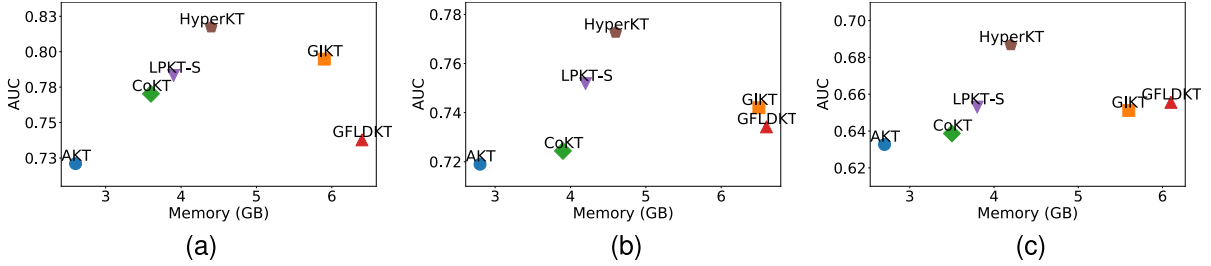


Fig. 8. Comparison of memory consumption (GB) and AUC on all datasets. (a) Assist2009. (b) EdNet1. (c) EdNet2.

G. Efficiency Evaluation

To assess the model's time efficiency, we conduct a comparative analysis of the runtime consumption between HyperKT and several SOTA methods. Fig. 7 shows the runtime for models achieving the best performance on the test set (for clarity, AUC values are utilized to illustrate the model's performance). As observed, graph-based models GIKT, GFLDKT, and HyperKT display the longest runtimes while showing relatively robust performance. HyperKT maintains superior efficiency compared to GIKT and GFLDKT. Furthermore, despite AKT demonstrating the fastest convergence speed, its AUC value is significantly lower than that of HyperKT. In summary, the proposed HyperKT model achieves the best performance without significantly increasing the runtime cost.

To evaluate the model's efficiency in terms of memory consumption, we conducted a comparative analysis of memory usage and AUC values for models, including HyperKT and several SOTA methods. Fig. 8 illustrates the relationship between memory usage and the AUC values, providing insights into the trade-off between computational resources

and model performance. As observed, graph-based models GIKT and GFLDKT exhibit the highest memory consumption. AKT takes up the lowest memory but has the worst performance. HyperKT outperforms CoKT and LPKT-S, which have approximative memory consumption. In summary, HyperKT achieves the best performance without compromising the efficiency of model memory usage.

VI. CONCLUSION

In this article, we propose a novel KT model HyperKT, which goes beyond limited observed pairwise links and captures complex high-order information via hypergraph learning. We design adaptive scale hypergraph distillation to automatically generate significant hyperedges. We propose dual-channel hypergraph encoders to refine the multigranularity knowledge states of learners, effectively transforming the higher-order information in the constructed state hypergraph. Introduced cross-view contrastive learning maximizes the mutual information of multilevel knowledge state, guiding solving the problems of deficient hypergraph supervisory

signal and data sparse. Our extensive experiments on three real-world datasets have validated the effectiveness of the proposed HyperKT and each of its modules. Overall, the HyperKT model based on hypergraphs provides a novel paradigm for predicting learner performance.

REFERENCES

- [1] C. Conati, O. Barral, V. Putnam, and L. Rieger, "Toward personalized XAI: A case study in intelligent tutoring systems," *Artif. Intell.*, vol. 298, Sep. 2021, Art. no. 103503.
- [2] F. Wang et al., "NeuralCD: A general framework for cognitive diagnosis," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 8, pp. 8312–8327, Aug. 2023.
- [3] Z. Liu, Q. Liu, J. Chen, S. Huang, and W. Luo, "simpleKT: A simple but tough-to-beat baseline for knowledge tracing," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023, pp. 1–13.
- [4] S. An, J. Kim, M. Kim, and J. Park, "No task left behind: Multi-task learning of knowledge tracing and option tracing for better student assessment," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, pp. 4424–4431.
- [5] S. Minn, J.-J. Vie, K. Takeuchi, H. Kashima, and F. Zhu, "Interpretable knowledge tracing: Simple and efficient student modeling with causal relations," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2022, pp. 12810–12818.
- [6] Q. Liu et al., "EKT: Exercise-aware knowledge tracing for student performance prediction," *IEEE Trans. Knowl. Data Eng.*, vol. 33, no. 1, pp. 100–115, Jan. 2021.
- [7] F. Liu, X. Hu, C. Bu, and K. Yu, "Fuzzy Bayesian knowledge tracing," *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 7, pp. 2412–2425, Jul. 2022.
- [8] J. Vie and H. Kashima, "Knowledge tracing machines: Factorization machines for knowledge tracing," in *Proc. AAAI Conf. Artif. Intell.*, 2019, pp. 750–757.
- [9] A. Ghosh, N. Heffernan, and A. S. Lan, "Context-aware attentive knowledge tracing," in *Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2020, pp. 2330–2339.
- [10] Z. Liu, Q. Liu, J. Chen, S. Huang, J. Tang, and W. Luo, "pyKT: A Python library to benchmark deep learning based knowledge tracing models," in *Proc. Conf. Neural Inf. Process. Syst. (NIPS)*, vol. 35, 2022, pp. 18542–18555.
- [11] C. Piech et al., "Deep knowledge tracing," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 505–513.
- [12] X. Sun, X. Zhao, B. Li, Y. Ma, R. Sutcliffe, and J. Feng, "Dynamic key-value memory networks with rich features for knowledge tracing," *IEEE Trans. Cybern.*, vol. 52, no. 8, pp. 8239–8245, Aug. 2022.
- [13] K. Nagatani, Q. Zhang, M. Sato, Y.-Y. Chen, F. Chen, and T. Ohkuma, "Augmenting knowledge tracing by considering forgetting behavior," in *Proc. World Wide Web Conf.*, May 2019, pp. 3101–3107.
- [14] S. Shen et al., "Monitoring student progress for learning process-consistent knowledge tracing," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 8, pp. 8213–8227, Aug. 2023.
- [15] T. Long et al., "Improving knowledge tracing with collaborative information," in *Proc. 15th ACM Int. Conf. Web Search Data Mining*, Feb. 2022, pp. 599–607.
- [16] C. Wang et al., "Temporal cross-effects in knowledge tracing," in *Proc. 14th ACM Int. Conf. Web Search Data Mining*, Mar. 2021, pp. 517–525.
- [17] X. Song, J. Li, Y. Tang, T. Zhao, Y. Chen, and Z. Guan, "JKT: A joint graph convolutional network based deep knowledge tracing," *Inf. Sci.*, vol. 580, pp. 510–523, Nov. 2021.
- [18] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and S. Y. Philip, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, Mar. 2020.
- [19] Y. Yang et al., "GIKT: A graph-based interaction model for knowledge tracing," in *Proc. ECML PKDD*, 2020, pp. 299–315.
- [20] W. Zhao, J. Xia, X. Jiang, and T. He, "A novel framework for deep knowledge tracing via gating-controlled forgetting and learning mechanisms," *Inf. Process. Manag.*, vol. 60, no. 1, Jan. 2023, Art. no. 103114.
- [21] H. Tong, Z. Wang, Y. Zhou, S. Tong, W. Han, and Q. Liu, "Introducing problem schema with hierarchical exercise graph for knowledge tracing," in *Proc. 45th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2022, pp. 405–415.
- [22] C. Zhang, S. Hu, Z. G. Tang, and T.-H. H. Chan, "Re-visiting learning on hypergraphs: Confidence interval, subgradient method, and extension to multiclass," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 3, pp. 506–518, Mar. 2020.
- [23] V. La Gatta, V. Moscato, M. Pennone, M. Postiglione, and G. Sperli, "Music recommendation via hypergraph embedding," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7887–7899, Oct. 2023.
- [24] Y. Gao, Z. Zhang, H. Lin, X. Zhao, S. Du, and C. Zou, "Hypergraph learning: Methods and practices," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2548–2566, May 2022.
- [25] X. Sun et al., "Multi-level hyperedge distillation for social linking prediction on sparsely observed networks," in *Proc. Int. Conf. World Wide Web. (WWW)*, 2021, pp. 2934–2945.
- [26] G. Abdelrahman, Q. Wang, and B. P. Nunes, "Knowledge tracing: A survey," *ACM Comput. Surv.*, vol. 55, no. 11, pp. 1–37, 1145.
- [27] A. T. Corbett and J. R. Anderson, "Knowledge tracing: Modeling the acquisition of procedural knowledge," *User Model. User Adapt. Interact.*, vol. 4, no. 4, pp. 253–278, 1994.
- [28] T. Käser, S. Klingler, A. G. Schwing, and M. Gross, "Dynamic Bayesian networks for student modeling," *IEEE Trans. Learn. Technol.*, vol. 10, no. 4, pp. 450–462, Oct. 2017.
- [29] M. Zhang, X. Zhu, C. Zhang, Y. Ji, F. Pan, and C. Yin, "Multi-factors aware dual-attentional knowledge tracing," in *Proc. 30th ACM Int. Conf. Inf. Knowl. Manag.*, Oct. 2021, pp. 2588–2597.
- [30] S. Pandey and G. Karypis, "A self attentive model for knowledge tracing," in *Proc. 12th Int. Conf. Educ. Data Min. (EDM)*, 2019, pp. 384–389.
- [31] T. Long, Y. Liu, J. Shen, W. Zhang, and Y. Yu, "Tracing knowledge state with individual cognition and acquisition estimation," in *Proc. 44th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2021, pp. 173–182.
- [32] Y. Liu, Y. Yang, X. Chen, J. Shen, H. Zhang, and Y. Yu, "Improving knowledge tracing via pre-training question embeddings," in *Proc. 29th Int. Joint Conf. Artif. Intell.*, Jul. 2020, pp. 1577–1583.
- [33] N. Yadati, M. Nimishakavi, P. Yadav, V. Nitin, A. Louis, and P. P. Talukdar, "HyperGCN: A new method for training graph convolutional networks on hypergraphs," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2019, pp. 1509–1520.
- [34] L. Sun, Y. Rao, X. Zhang, Y. Lan, and S. Yu, "MS-HGAT: Memory-enhanced sequential hypergraph attention network for information diffusion prediction," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 4156–4164.
- [35] D. A. Nguyen, C. H. Nguyen, and H. Mamitsuka, "Central-smoothing hypergraph neural networks for predicting drug–drug interactions," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 3, 2023, doi: 10.1109/TNNLS.2023.3261860.
- [36] Y. Gao, M. Wang, D. Tao, R. Ji, and Q. Dai, "3-D object retrieval and recognition with hypergraph analysis," *IEEE Trans. Image Process.*, vol. 21, no. 9, pp. 4290–4303, Sep. 2012.
- [37] Y. Su, Y. Gu, Z. Wang, Y. Zhang, J. Qin, and G. Yu, "Efficient subhypergraph matching based on hyperedge features," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 5808–5822, Jun. 2023.
- [38] X. Xia, H. Yin, J. Yu, Q. Wang, L. Cui, and X. Zhang, "Self-supervised hypergraph convolutional networks for session-based recommendation," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 4503–4511.
- [39] Y. Li, C. Gao, H. Luo, D. Jin, and Y. Li, "Enhancing hypergraph neural networks with intent disentanglement for session-based recommendation," in *Proc. 45th Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2022, pp. 1997–2002.
- [40] J. Li, B. Liang, X. Lu, M. Li, G. Lu, and Y. Xu, "From global to local: Multi-patch and multi-scale contrastive similarity learning for unsupervised defocus blur detection," *IEEE Trans. Image Process.*, vol. 32, pp. 1158–1169, 2023.
- [41] Y. Liu, Z. Li, S. Pan, C. Gong, C. Zhou, and G. Karypis, "Anomaly detection on attributed networks via contrastive self-supervised learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 6, pp. 2378–2392, Jun. 2022.
- [42] L. Peng et al., "GRLC: Graph representation learning with constraints," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 3, 2023, doi: 10.1109/TNNLS.2022.3230979.
- [43] L. Wu, H. Lin, C. Tan, Z. Gao, and S. Z. Li, "Self-supervised learning on graphs: Contrastive, generative, or predictive," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 4216–4235, Apr. 2023.
- [44] Y. You, T. Chen, Y. Sui, T. Chen, Z. Wang, and Y. Shen, "Graph contrastive learning with augmentations," in *Proc. NIPS*, vol. 33, 2020, pp. 5812–5823.

- [45] W. Lee, J. Chun, Y. Lee, K. Park, and S. Park, "Contrastive learning for knowledge tracing," in *Proc. Int. Conf. World Wide Web. (WWW)*, 2022, pp. 2330–2338.
- [46] X. Song, J. Li, Q. Lei, W. Zhao, Y. Chen, and A. Mian, "Bi-CLKT: Bi-graph contrastive learning based knowledge tracing," *Knowl.-Based Syst.*, vol. 241, Apr. 2022, Art. no. 108274.
- [47] S. Wan et al., "Boosting graph contrastive learning via adaptive sampling," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jul. 13, 2023, doi: [10.1109/TNNLS.2023.3291358](https://doi.org/10.1109/TNNLS.2023.3291358).
- [48] H. Zhao, X. Yang, C. Deng, and D. Tao, "Unsupervised structure-adaptive graph contrastive learning," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 5, 2023, doi: [10.1109/TNNLS.2023.3271140](https://doi.org/10.1109/TNNLS.2023.3271140).
- [49] W. Zhuo and G. Tan, "Graph contrastive learning with adaptive proximity-based graph augmentation," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 5, 2023, doi: [10.1109/TNNLS.2023.3278183](https://doi.org/10.1109/TNNLS.2023.3278183).
- [50] X. He, K. Deng, X. Wang, Y. Li, Y. Zhang, and M. Wang, "LightGCN: Simplifying and powering graph convolution network for recommendation," in *Proc. 43rd Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, Jul. 2020, pp. 639–648.
- [51] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
- [52] J. Ma, M. Wan, L. Yang, J. Li, B. Hecht, and J. Teevan, "Learning causal effects on hypergraphs," in *Proc. ACM SIGKDD Int. Conf. Knowl. Disc. Data Min. (KDD)*, 2022, pp. 1202–1212.



Jiawei Li received the B.S. degree in software engineering from Shandong Jianzhu University, Jinan, China, in 2018. He is currently pursuing the Ph.D. degree in software engineering with South China Normal University, Guangzhou, China.

He has authored papers in referred journals and conferences, such as IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS (TCSS), IEEE TRANSACTIONS ON BIG DATA (TBD), and International

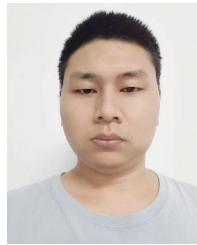
Conference on Knowledge Science, Engineering and Management (KSEM). His research interests include deep learning, recommendation systems, and artificial intelligence in education.



Yuanfei Deng received the Ph.D. degree in theory and technology of service computing from South China Normal University, Guangzhou, China, in 2023.

In 2019, he was approved by the "International Training Project for Outstanding Young Scientific Research Talents in Guangdong Universities" to study at Carleton University, Ottawa, ON, Canada, as a Joint Training Graduate. His work is published in journals and conferences, such as IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING

SYSTEMS (TNNLS), IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS (TCSS), Knowledge-Based Systems (KBS), and International Conference on Computer Supported Cooperative Work in Design (CSCWD). His current research interests include semantic computing, knowledge graph, and data science.



Yixiu Qin received the M.S. degree in computer science and technology from Guilin University of Electronic Science and Technology, Guilin, China, in 2018. He is currently pursuing the Ph.D. degree in software engineering with South China Normal University, Guangzhou, China.

His research has been published in journals, including IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS (TCSS), and IEEE TRANSACTIONS ON BIG DATA (TBD). His research interests include deep learning, multilabel classification, and fine-grained entity typing.



Shun Mao received the M.S. degree in software engineering from South China Normal University, Guangzhou, China, in 2019, where he is currently pursuing the Ph.D. degree in software engineering.

His research has been published in journals and conferences, including IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS (TCSS), Information Processing and Management (IPM), IEEE TRANSACTIONS ON LEARNING TECHNOLOGIES (TLT), International Conference on Knowledge Science, Engineering and Management (KSEM), and Pacific Rim International Conference on Artificial Intelligence (PRICAI). His research interests include machine learning, learning analysis, and artificial intelligence in education.



Yuncheng Jiang (Member, IEEE) received the Ph.D. degree from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2004.

He is currently a Professor at the School of Computer Science, the School of Artificial Intelligence, and the School of Software, South China Normal University, Guangzhou, China. His research has been published in journals and conferences, including IEEE TRANSACTIONS ON KNOWLEDGE AND DATA ENGINEERING (TKDE), IEEE TRANSACTIONS ON

NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), Information Processing and Management (IPM), *Neural Networks*, *Information Sciences*, *Fuzzy Sets and Systems*, IEEE TRANSACTIONS ON LEARNING TECHNOLOGIES (TLT), IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS (TCSS), IEEE TRANSACTIONS ON BIG DATA (TBD), Knowledge And Information Systems (KAIS), Knowledge-Based Systems (KBS), Engineering Applications of Artificial Intelligence (EAAI), International Journal of Intelligent Systems (IJIS), Association for the Advancement of Artificial Intelligence Conference on Artificial Intelligence (AAAI), International Joint Conference on Artificial Intelligence (IJCAI), Conference on Information and Knowledge Management (CIKM), and The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD), and International Conference on Knowledge Science, Engineering and Management (KSEM). His current research interests include graph computing, knowledge graphs, graph neural networks, educational artificial intelligence, and data science.

Dr. Jiang is a member of KSEM Steering Committee, the Vice Chair for the IEEE Task Force on Educational Data Mining, and a Distinguished Member of the China Computer Federation (CCF). He was a recipient of the Best Paper Award at KSEM 2022. He is also the General Co-Chair for CCF NCTCS 2023, the PC Co-Chair for KSEM 2023 and CAAI CDIC 2023, and the Workshop Chair for Pacific Rim International Conference on Artificial Intelligence (PRICAI) 2024.